

1234567891011

CONVERTIDOR BACK-TO-BACK (AFE + INVERSOR TRIFASICO)

Flujo de energia: RED -> [01 LCL] -> [02 AFE] == [03 BUS DC] == [04 INVERSOR] -> MOTOR
Frenado regenerativo: el AFE devuelve energia a la red; el chopper de [03] es solo respaldo.
Control [07 STM32] con proteccion por hardware [08]. Telemetria [11 ESP32] tras barrera de aislamiento [10].

Las hojas se conectan por GLOBAL LABELS (DC_P, DC_N, SW_*, G_*, K_*, PWM_*, AIN_*, FAULT_N, ...).

01_AC_Input_LCL

02_AFE_Bridge

03_DClink_Precharge

04_Inverter_Bridge

05_Gate_Drivers

06_Sense_LV

07_MCU_STM32

08_Protection_IO

09_PSU_Aux_Bias

10_Isolation_Comms

11_ESP32_IoT

Reglas DRC: kicad/drc_custom_rules.txt
Generado por tools/gen_schematic.py
AFE + Inversor back-to-back | STM32G474 | ESP32-S3
15 kVA | 400 VLL 60 Hz | Vdc <= 800 V | fsw 16 kHz
SistemasControlLAPAH

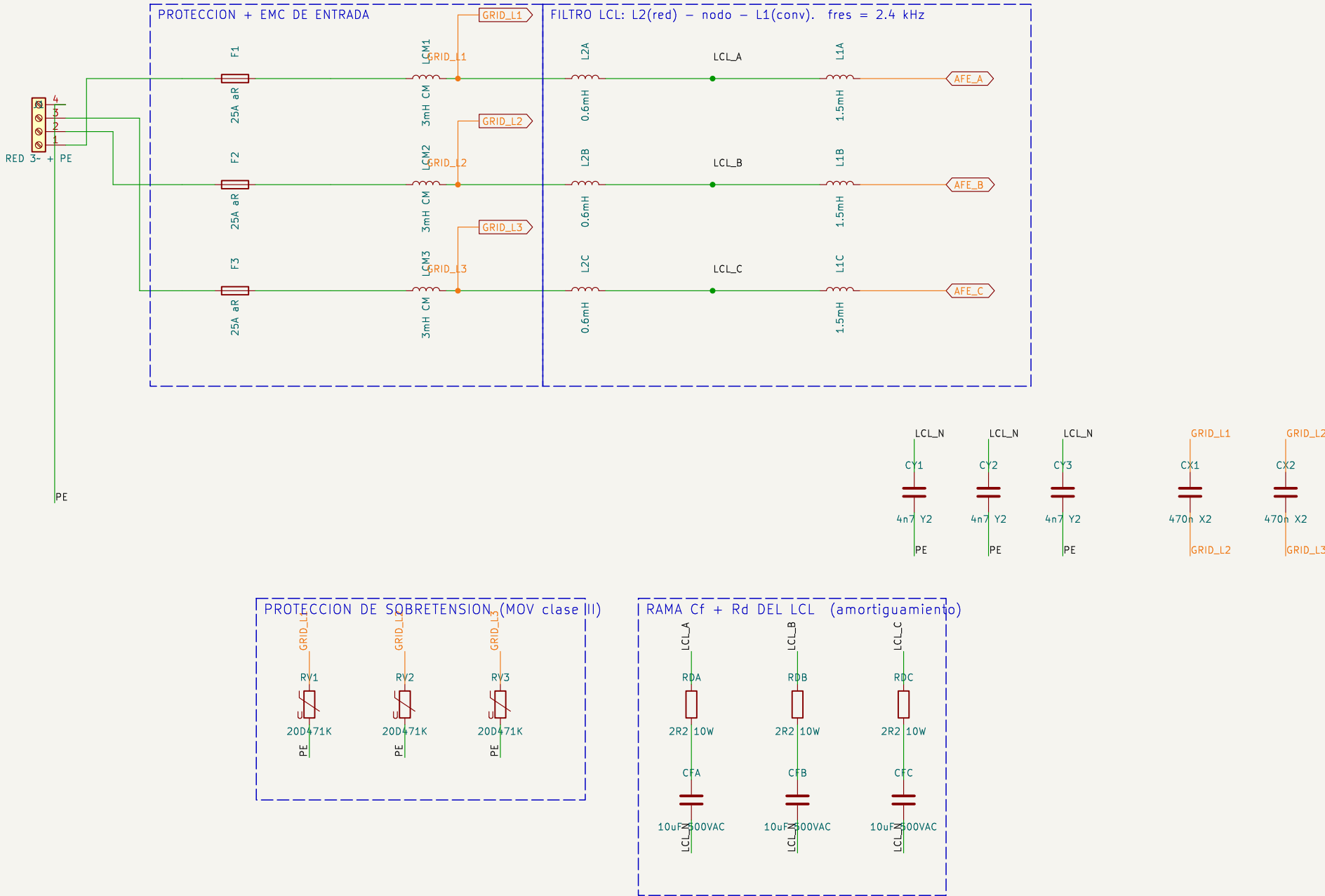
Sheet: /
File: b2b_converter.kicad_sch

Title: B2B CONVERTER -- HOJA RAIZ

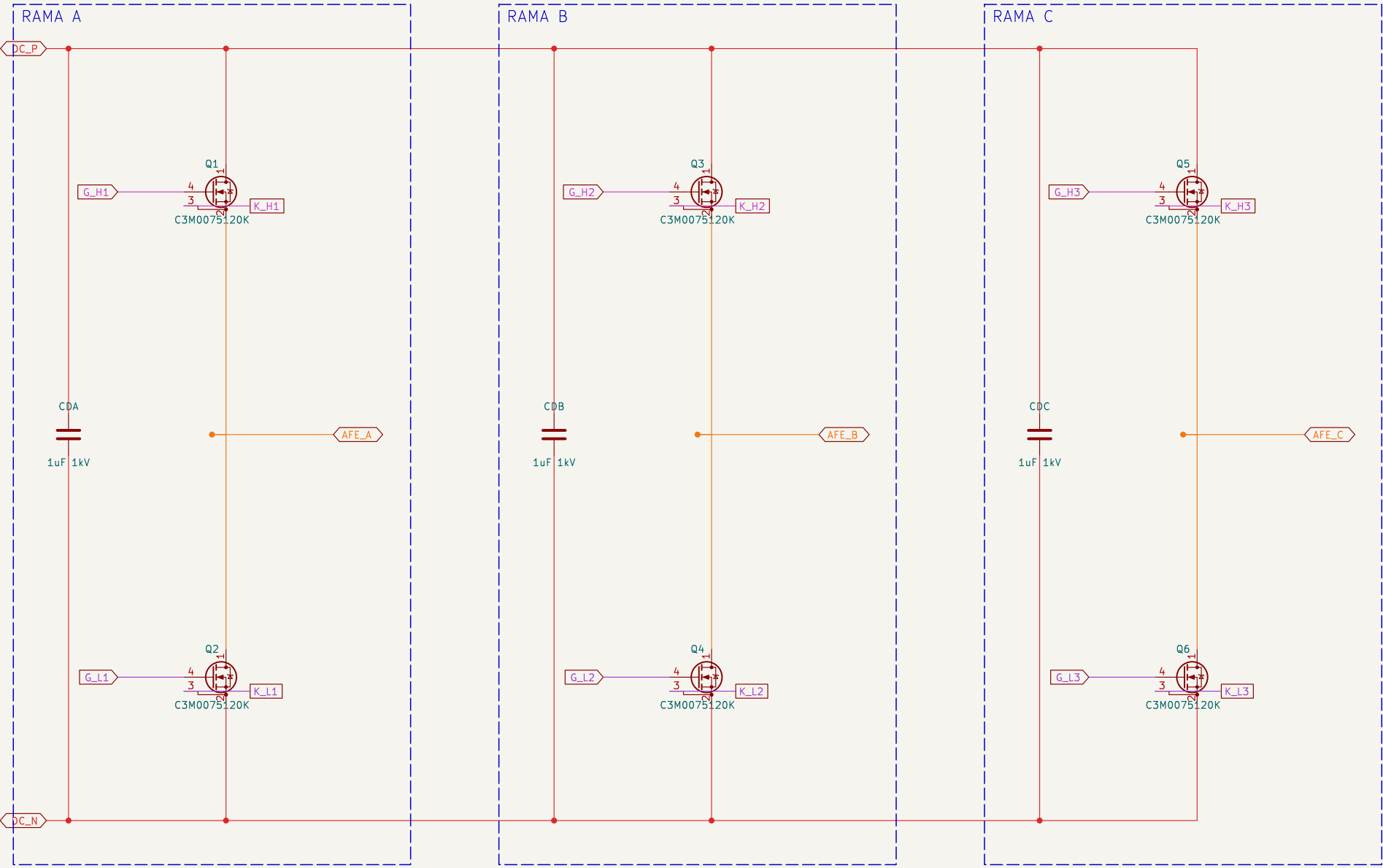
Size: A2 | Date: 2026-09-05 | Rev: A
KiCad E.D.A. 9.0.2 | Id: 1/12

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SHEET 01 – ENTRADA AC + FILTRO LCL
Orden físico: RED -> fusible -> contactor -> choke CM -> L2(red) -> [Cf+Rd al neutro] -> L1(conv) -> puente AFE
Clase de red: HV_AC (clearance 3.0 mm / creepage 4.0 mm)

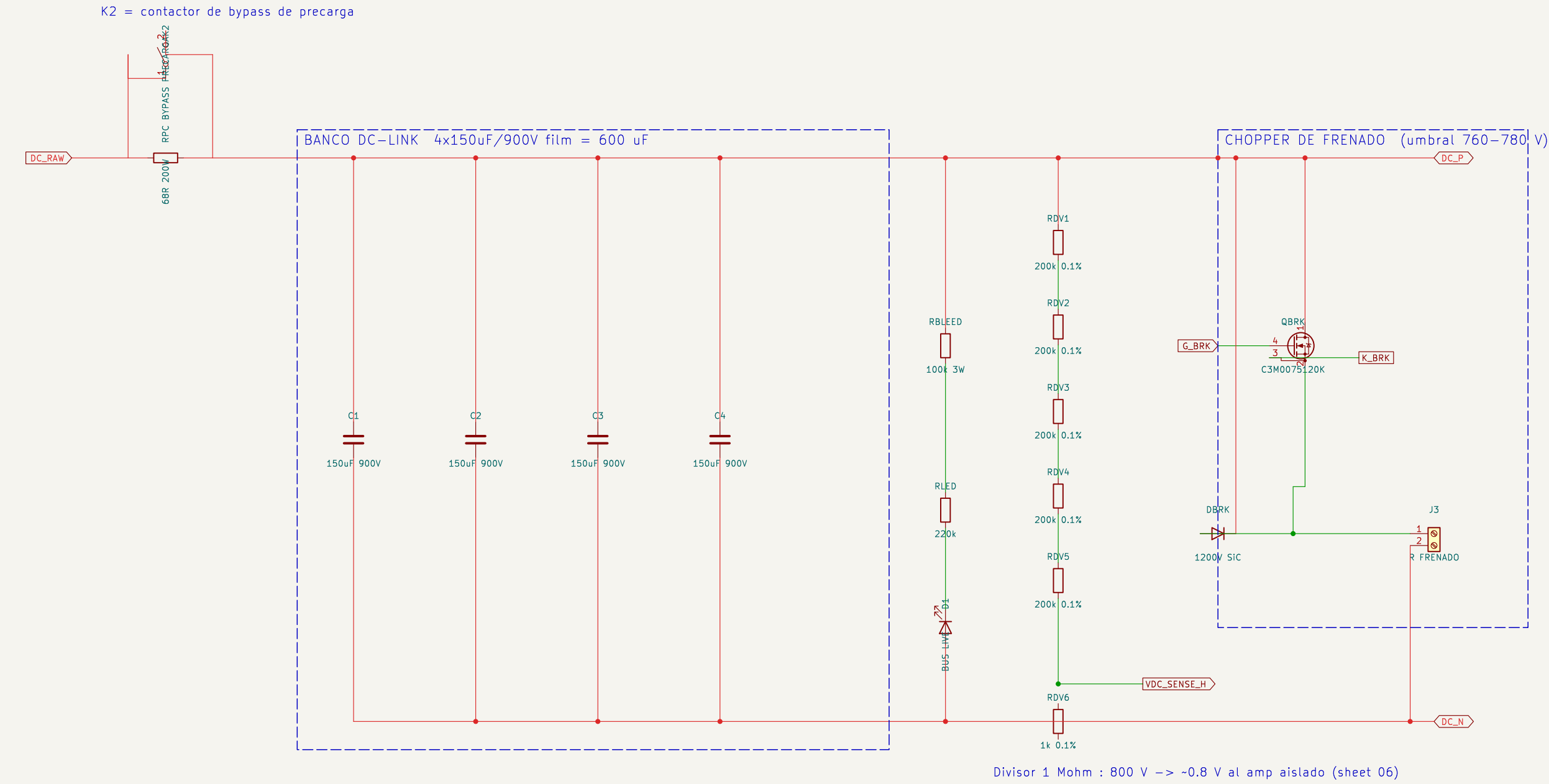


SHEET 02 – PUENTE AFE
Ramas A/B/C. El nodo de conmutacion ES la salida del LCL (AFE_x).
Gates G_H1..3 / G_L1..3 y Kelvin K_H1..3 / K_L1..3 al sheet 05.



LAZO DE CONMUTACION: el condensador de 1 uF/1 kV va a menos de 10 mm del par de MOSFET. Objetivo L_lazo < 20 nH.

SHEET 03 – BUS DC / PRECARGA / CHOPPER DE FRENADO
Precarga: Rpc 68 ohm limita el inrush a 8 A. tau = 48 ms. K2 puentea a Vdc >= 510 V o t = 300 ms.
Energía por carga en Rpc = 112 J -> resistencia bobinada con rating de pulso.



Banco C_dc 600 uF / 900 V – precarga 68 ohm – chopper de frenado

SistemasControlPAPUH

Sheet: /03_DClink_Precarga/

File: dclink_precarga.kicad_sch

Title: 03 – Bus DC + Precarga + Chopper

Size: A2

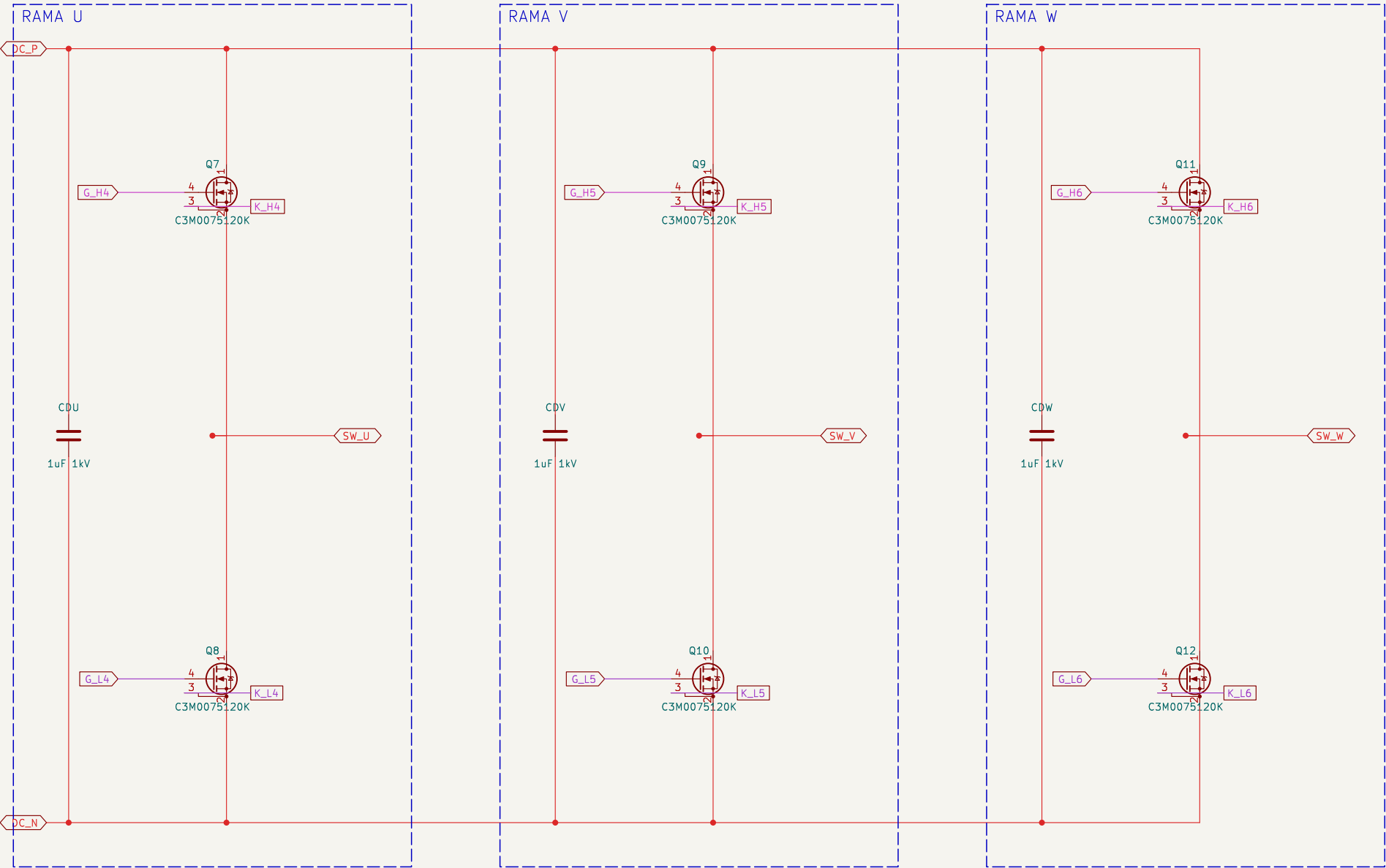
Date: 2026-09-05

Rev: A

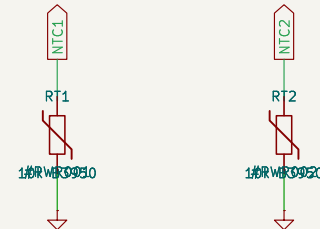
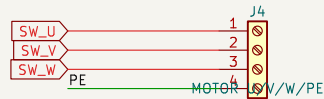
KiCad E.D.A. 9.0.2

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SHEET 04 – PUENTE INVERSOR
Ramas U/V/W. Gates G_H4..6 / G_L4..6 y Kelvin al sheet 05.
Salida a bornera de motor con PE y abrazadera de malla 360 grados.



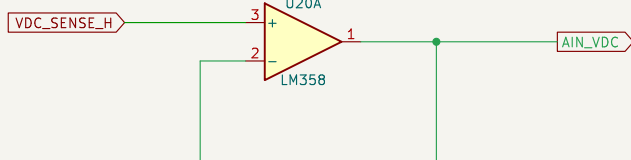
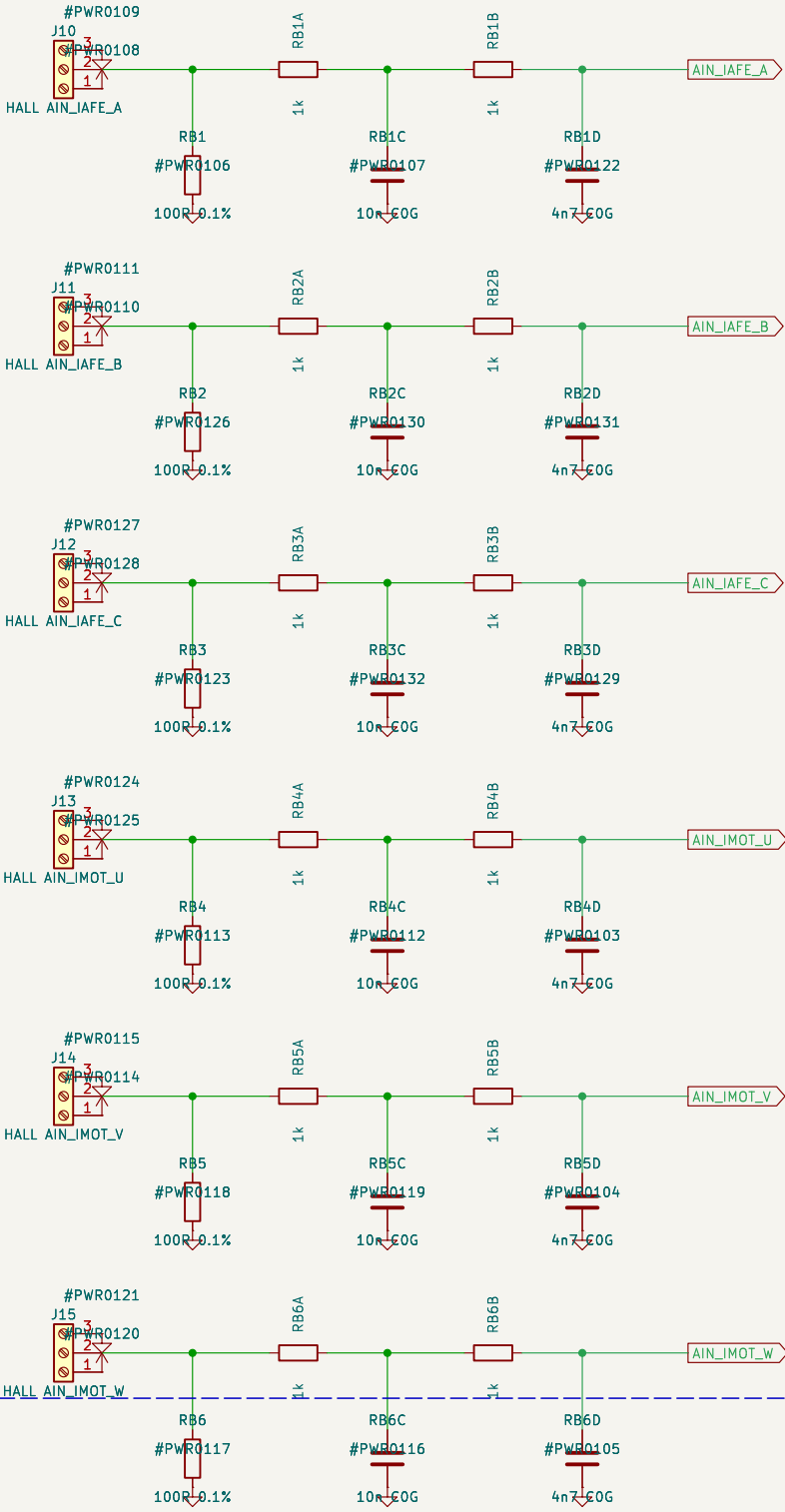
LAZO DE CONMUTACION: el condensador de 1 uF/1 kV va a menos de 10 mm del par de MOSFET. Objetivo L_lazo < 20 nH.



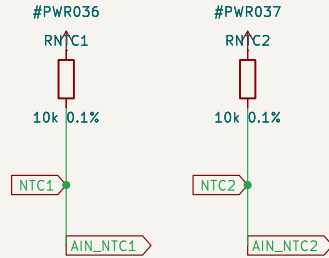
RT1 = disipador RT2 = cuerpo de modulo

SHEET 06 – SENSADO DE CORRIENTE Y TENSION
Cada canal: transductor Hall aislado -> resistencia burden 0.1% -> filtro anti-alias RC de 2 polos -> ADC del STM32.
El muestreo va sincronizado al centro del PWM, por eso fc puede ser ~14 kHz sin degradar el lazo (retardo de grupo < 1 us).

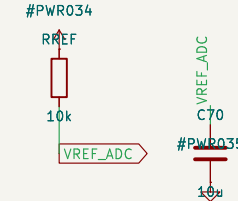
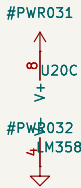
CANALES DE CORRIENTE (6)



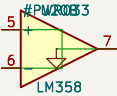
Buffer seguidor del divisor de bus (sheet 03)



Divisores de NTC (disipador y modulo)

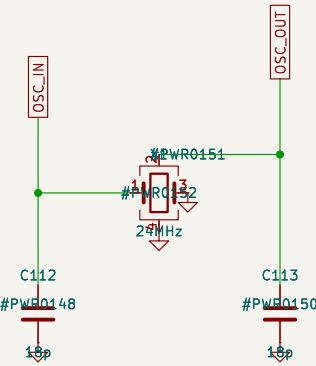
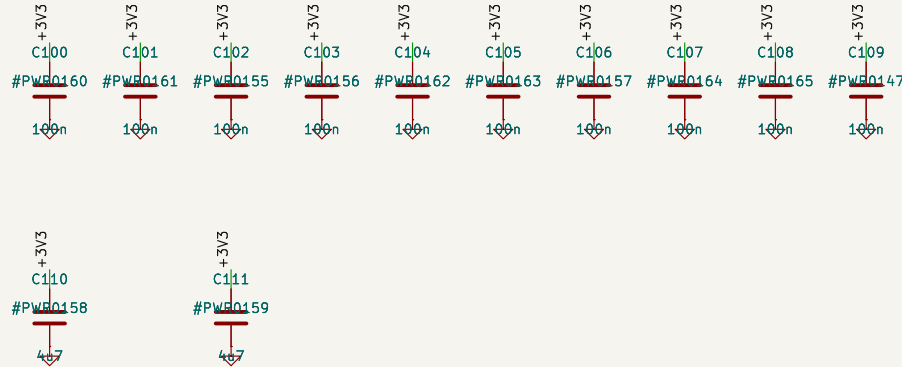
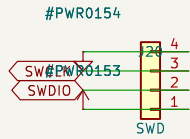
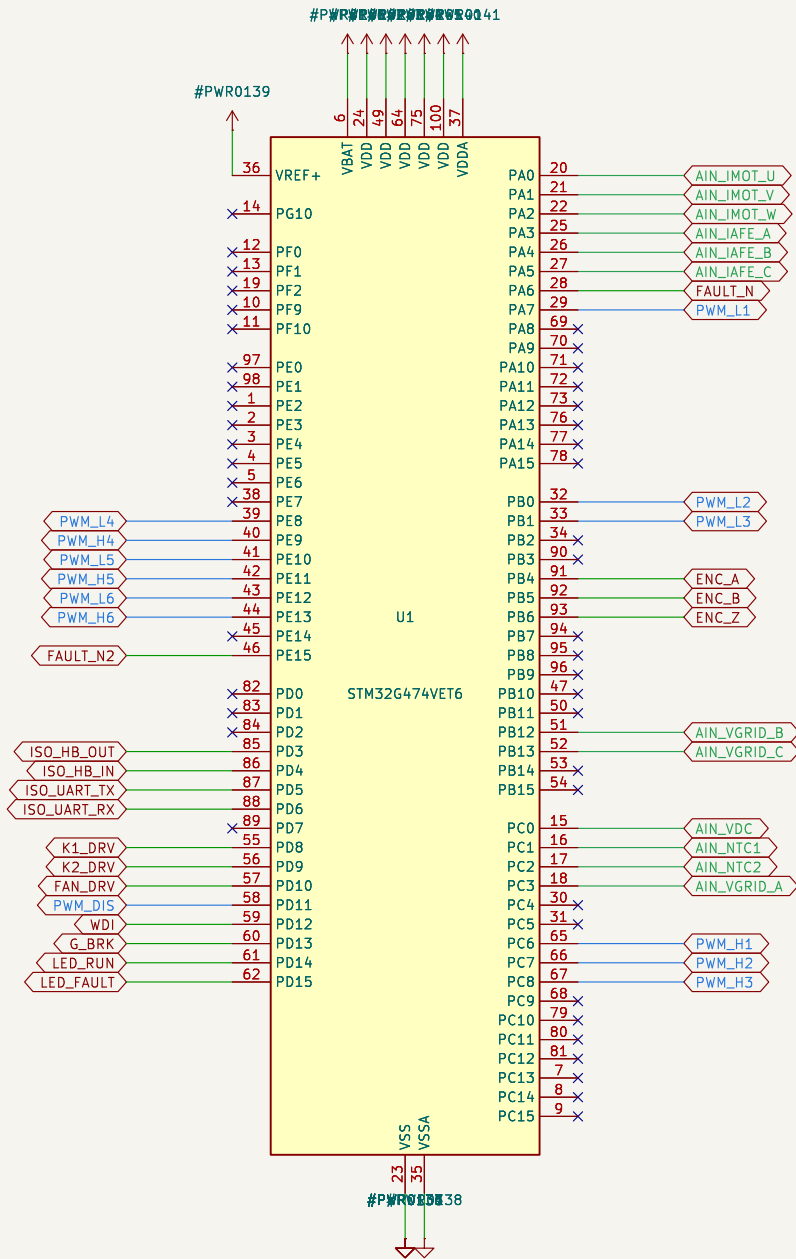


VREF_ADC: sustituir por REF5030 en la version final.
Aqui se deja el nodo creado para el ruteo en estrella al pin VREF+ del STM32.



U20B: unidad libre configurada como seguidor.

SHEET 07 - STM32G474VET6 (LQFP-100)
TIM1 -> 6 PWM del inversor. TIM8 -> 6 PWM del AFE. BKIN/BKIN2 <- FAULT_N (apagado por hardware).
NT1 une AGND<->DGND bajo VREF-. NT2 une DGND<->PGND en el retorno del sensado de DC_N.



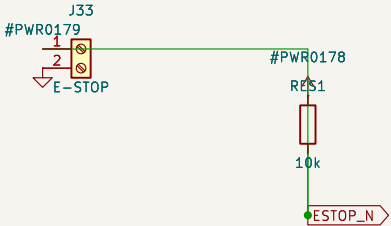
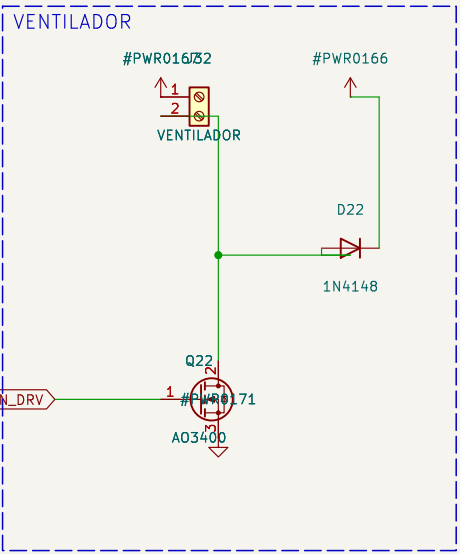
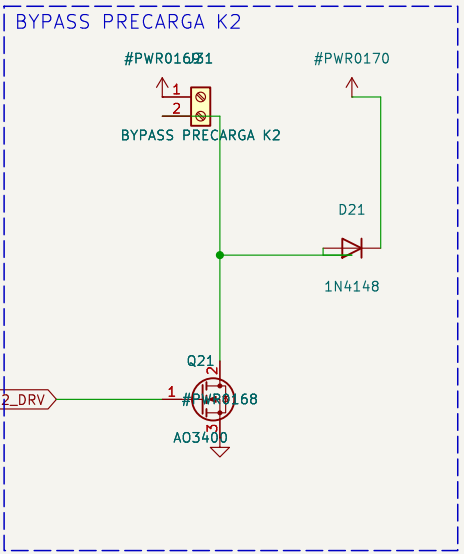
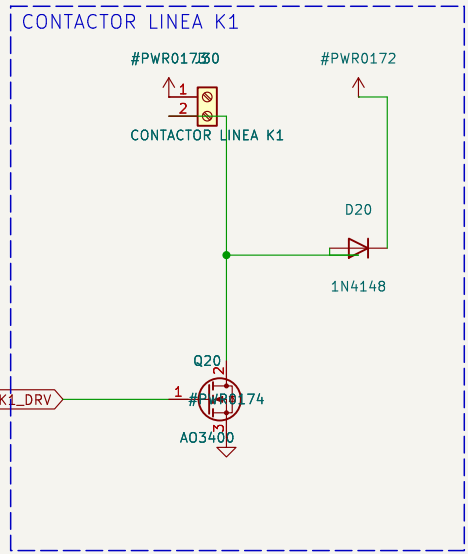
NT1 / NT2: union de tierras en UN SOLO punto.
IOGND (ESP32) NO se une aquí: es barrera.

CORDIC/FMAC para FOC y PLL trifasico
TIM1/TIM8 SVPWM con dead-time HW, ADC sinc. al centro del PWM
SistemasControlLAPAH
Sheet: /07_MCU_STM32/
File: mcu_stm32.kicad_sch

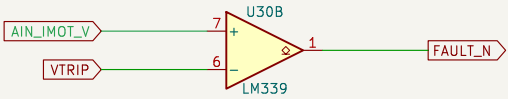
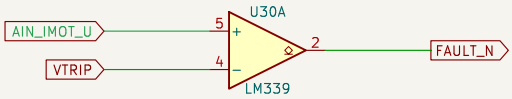
Title: 07 - MCU STM32G474VET6 (dominio de control)

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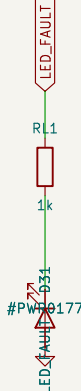
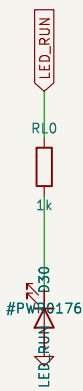
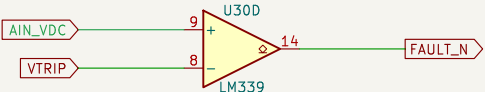
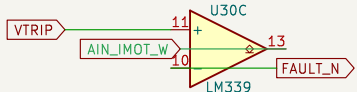
SHEET 08 – PROTECCION POR HARDWARE
La proteccion de sobrecorriente y shoot-through NO depende de software: comparador -> FAULT_N -> BKIN del timer -> MOE=0.
FAULT_N tambien apaga los 6 drivers via PWM_DIS y abre el contactor de linea K1.



E–STOP: contacto seco NC. Al abrir -> ESTOP_N alto -> falla.



U30 = 4 comparadores de disparo (open–collector, cableados en OR sobre FAULT_N con pull–up). VTRIP viene del divisor de umbral. Anadir histeresis con realimentacion positiva.



E–stop, rele K1/K2, watchdog, latch de falla
Cadena de falla independiente del firmware -> BKIN
SistemasControl_PAPUH

Sheet: /08_Proteccion_IO/
File: protection_io.kicad_sch

Title: 08 – Proteccion HW + I/O digital

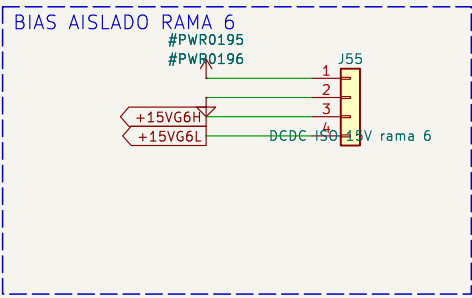
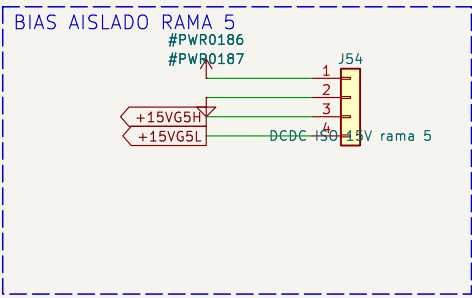
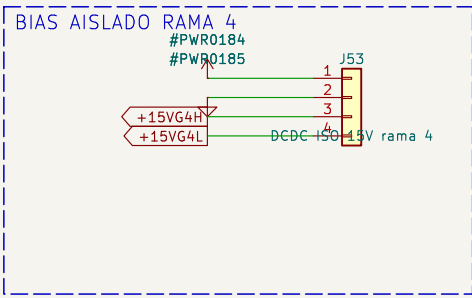
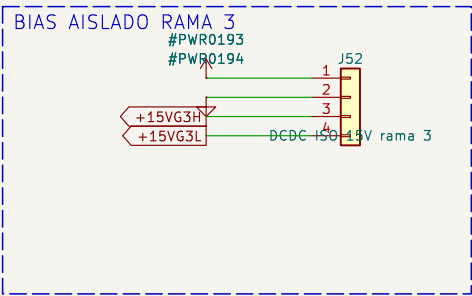
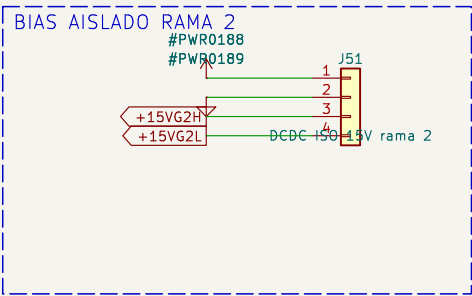
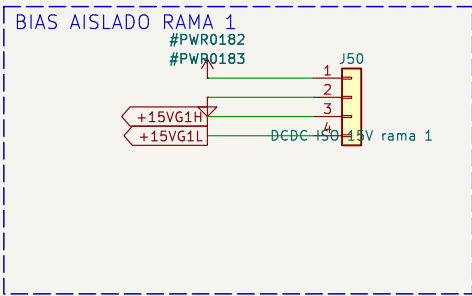
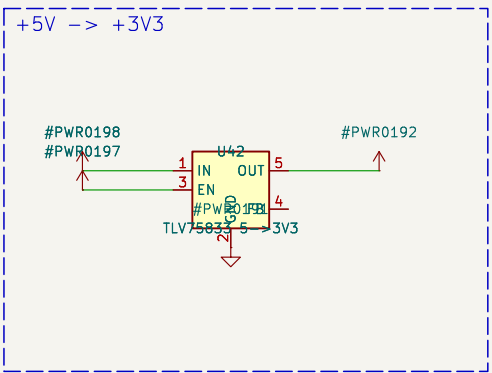
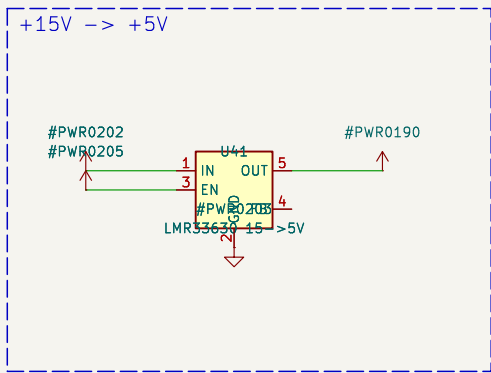
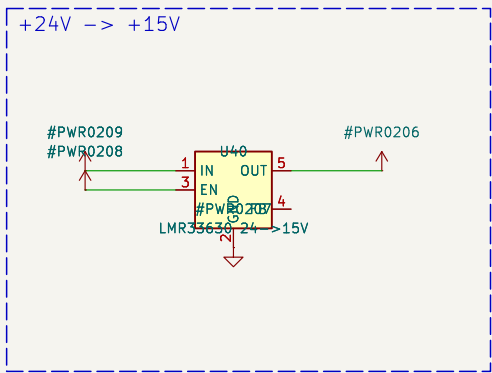
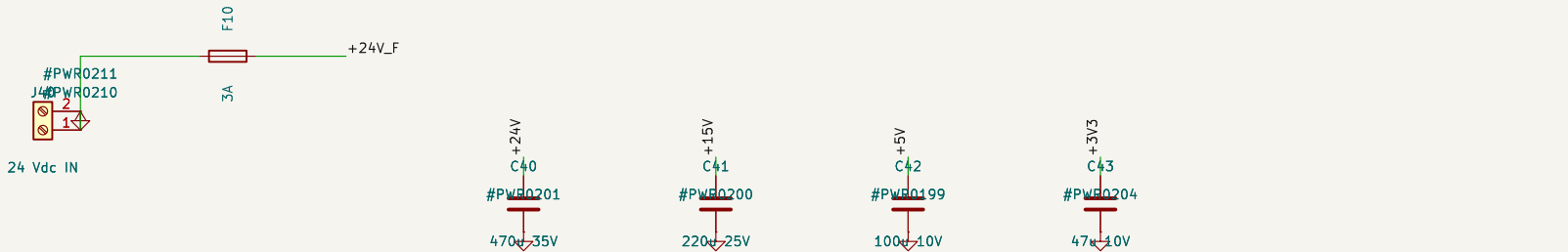
Size: A2 Date: 2026–09–05

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SHEET 09 – ALIMENTACION AUXILIAR
Entrada 24 Vdc externa (riel DIN). Rieles: +15 V (analógico y logica de driver), +5 V, +3V3.
Los 6 bias de gate son AISLADOS y flotan con el Kelvin de su rama. Sin ellos el driver no debe habilitarse (UVLO).



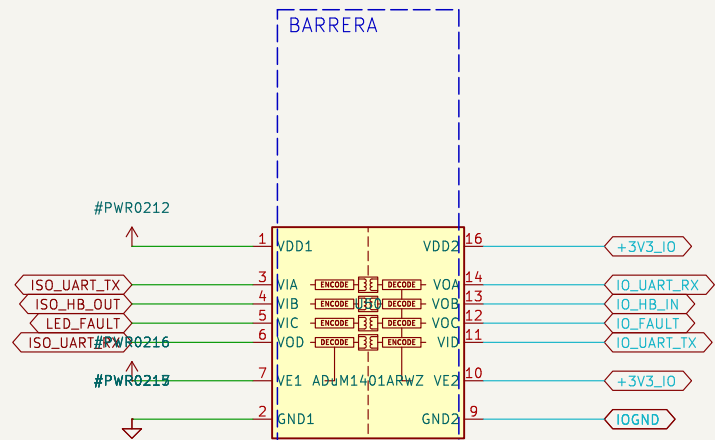
Cada modulo: 2 W, 15 V/−4 V, aislamiento reforzado.
Alternativa discreta: SN6505 + transformador + rectificador.

Secuenciado: bias antes de habilitar PWM
24 V → 15 V / 5 V / 3V3 + 6 bias aislados +15 V
SistemasControl_PAPUH
Sheet: /09_PSU_Aux_Bias/
File: psu_aux.kicad_sch

Title: 09 – Alimentacion auxiliar + bias aislado

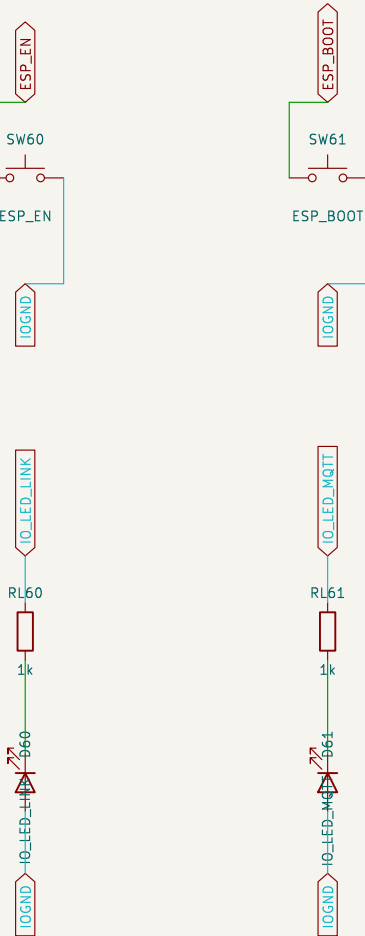
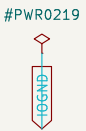
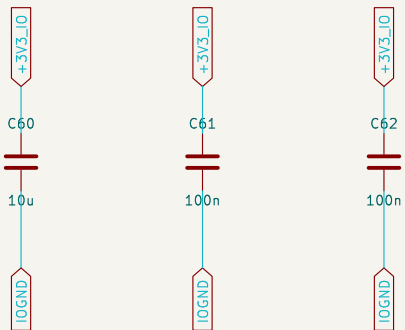
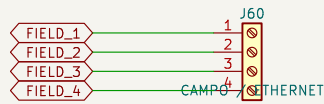
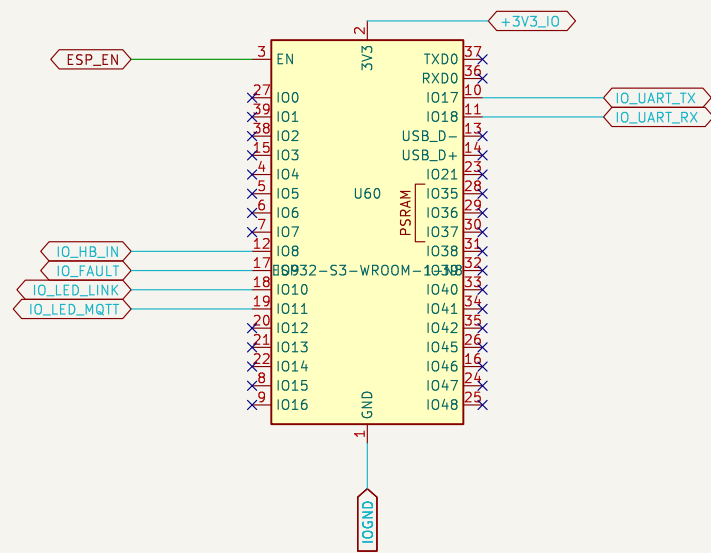
Size: A2 Date: 2026-09-05 Rev: A
KiCad E.D.A. 9.0.2 Id: 10/12

SHEET 10 – AISLAMIENTO DE COMUNICACIONES
Aislamiento FUNCIONAL (ambos dominios son SELV): mantiene el ruido de conmutacion y el rebote de masa fuera del ESP32.
GND1 = DGND del control. GND2 = IOGND del ESP32. NO se unen.
El ESP32 nunca esta en el lazo de par: solo telemetria y setpoints que el STM32 valida.



En la PCB: ranura fresada bajo U50 si el creepage del encapsulado es < 4 mm. Ninguna pista cruza esta línea.

SHEET 11 – ESP32–S3 (DOMINIO IoT AISLADO)
Solo telemetría. Recibe del STM32 por UART aislado y publica por MQTT/Modbus TCP.
KEEP–OUT de cobre bajo la antena del modulo (ver floorplan).



Modbus TCP + MQTT/TLS + OTA. Dominio IOGND aislado

SistemasControl_PAPUH

Sheet: /11_ESP32_IoT/

File: esp32_iot.kicad_sch

Title: 11 – ESP32–S3 pasarela IoT / SCADA

Size: A2

Date: 2026–09–05

Rev: A

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